

Robust Bin Packing with Budgeted Uncertainty

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- 1 Introduction to bin packing and robust optimization
- 2 Problem definition and uncertainty sets U^{Ω} and U^{Γ}
- 3 Complexity
- 4 Next-Cover-and-Repack approximations for U^{Ω} and U^{Γ}
- 5 Constant factor approximation for U^{Γ}
- 6 Conclusion

Classical bin packing



0.3



0.6



0.2



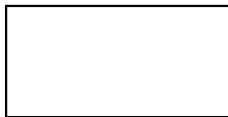
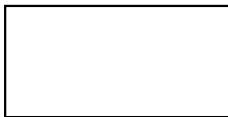
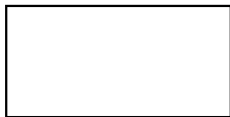
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Classical bin packing



0.3



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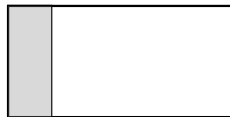
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Classical bin packing



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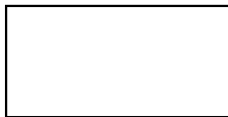
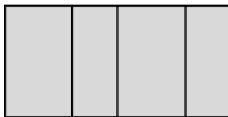
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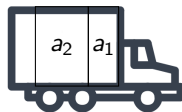
The classical bin packing problem

Standard bin packing is how to pack n items of given sizes into the least number of bins.

The Bin Packing Problem

Input: Items $a_1, \dots, a_n \in [0, 1]$

Output: A set of bins $B_1, \dots, B_{k^*} \subset 2^{[n]}$ of minimum cardinality k^* that feasibly partition the items: for each $j = 1, \dots, k^*$,
 $\sum_{i \in B_j} a_i \leq 1$ and $\bigcup_{j \in M^*} B_j = [n]$.



Classical bin packing approximation ratio results

Bound Type	Alg/LB	Value	Authors
Asymptotic	FFD	$11/9 \text{ OPT}$	Johnson (1973)
Asymptotic (FPTAS)	LP-based	$(1 + \epsilon) \text{ OPT}$	Karmakar & Karp (1982)
Absolute	LB	$3/2 \text{ OPT}$	
Absolute	FFD	$3/2 \text{ OPT}$	Simchi-Levi (1994)
Asymptotic	FFD	$11/9 \text{ OPT} + 7/9$	Dósa (2007)

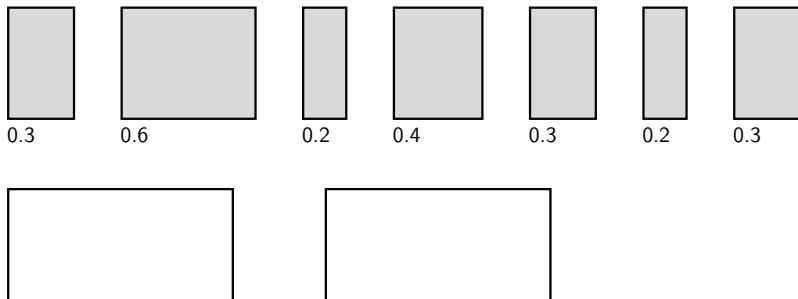
Recall: An asymptotic approximation ρ_∞ bound holds (in minimization) if

$$z(\mathcal{I}) \leq \rho_\infty z^*(\mathcal{I}) + \kappa,$$

for all $\mathcal{I}$, where κ is a constant with respect to the input size n .

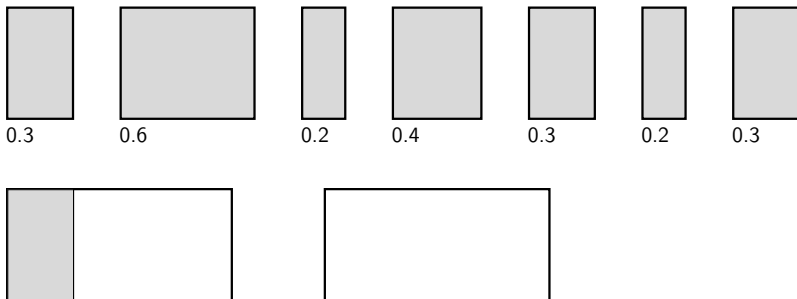
A simple bin packing algorithm

NEXT-COVER-AND-REPACK (NCR)



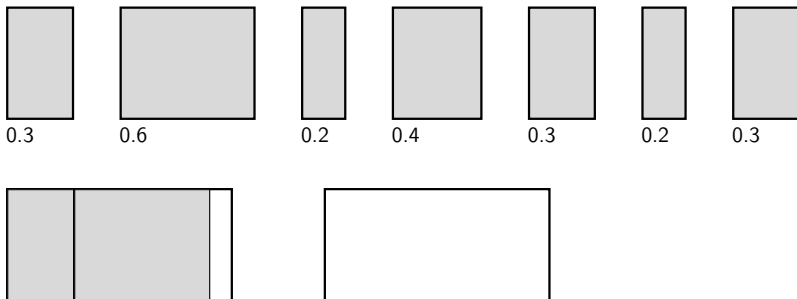
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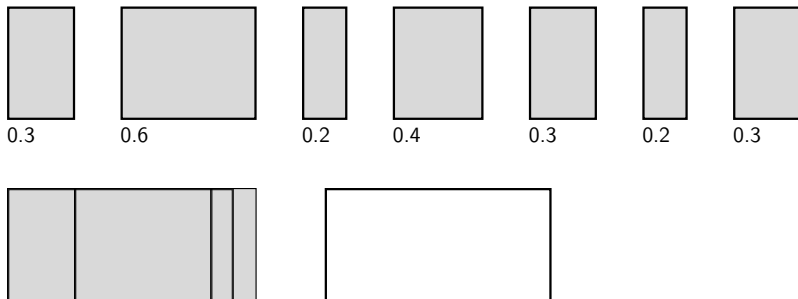
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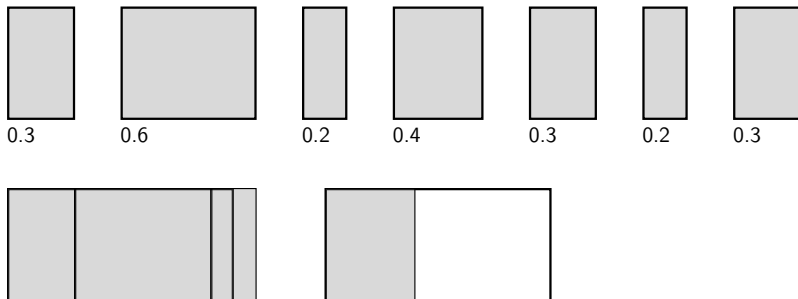
A simple bin packing algorithm

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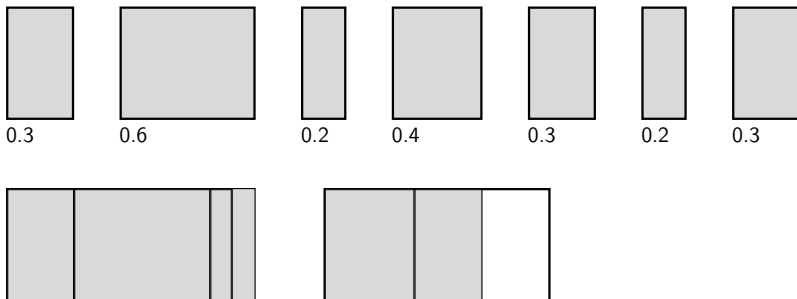
A simple bin packing algorithm

NEXT-COVER-AND-REPACK (NCR)



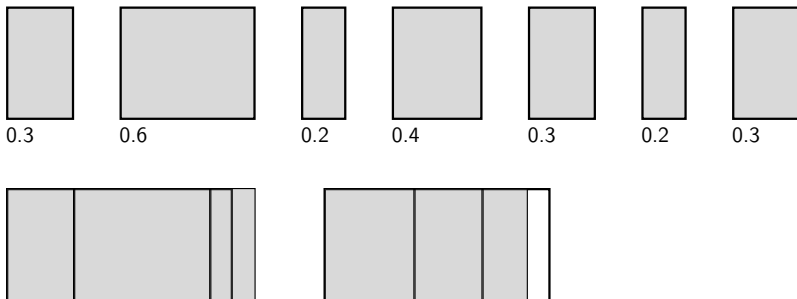
A simple bin packing algorithm

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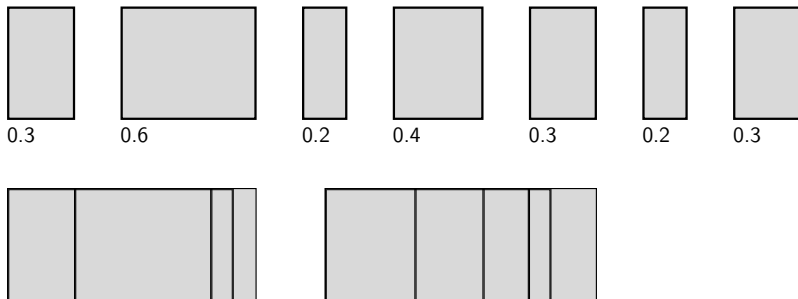
A simple bin packing algorithm

NEXT-COVER-AND-REPACK (NCR)



A simple bin packing algorithm

NEXT-COVER-AND-REPACK (NCR)



A simple bin packing algorithm



0.3



0.6



0.2



0.4



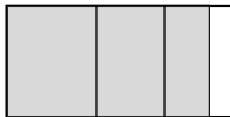
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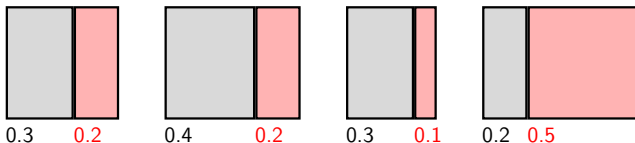
0.2



0.3



Practical motivation



1491

■ ECONOMICS

Anesthesiology
1999; 91:1491-500
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Lippincott Williams & Wilkins, Inc.

Which Algorithm for Scheduling Add-on Elective Cases Maximizes Operating Room Utilization?

*Use of Bin Packing Algorithms and Fuzzy Constraints in Operating
Room Management*

Franklin Dexter, M.D., Ph.D.,* Alex Macario, M.D., M.B.A.,† Rodney D. Traub, Ph.D.‡

Uncertainty sets in RO

Uncertainty	Definition of $U \subset \mathbb{R}^n$	Pros / Cons
Box	$\{a \mid l \leq a \leq u\}$	Overly conservative
Ellipsoid	$\{a \mid (a - \bar{a})^T P (a - \bar{a}) \leq 1\}$	Less conservative, probabilistic guarantees, intractable with binary decision variables
Polyhedral	$\{a \mid Pa \leq p\}$	Less conservative, tractable

See:

- Ben-Tal, El Ghaoui & Nemirovski (2009). Robust Optimization. Princeton University Press.
- Gorissen, Yanikoglu, den Hertog (2015). A Practical Guide to Robust Optimization.
- Buchheim & Kurtz (2018). Robust Combinatorial Optimization under Convex and Discrete Cost Uncertainty.

1 Introduction to bin packing and robust optimization

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The robust bin packing problem

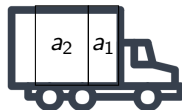
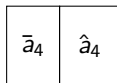
- Standard bin packing is how to pack items of given sizes into the least number of bins.
- In the robust bin packing that we consider items sizes are subject to uncertainty and can assume any value within an uncertainty set.

The Bin Packing Problem

Input:

Uncertainty set $U \subset [0, 1]^n$

Output: A set of bins $B_1, \dots, B_{k^*} \subset 2^{[n]}$ of minimum cardinality k^* that feasibly partition the items: for each $j = 1, \dots, k^*$,
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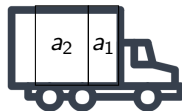
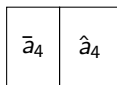
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The **Robust** Bin Packing Problem

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Uncertainty set $U \subset [0, 1]^n$

Output: A set of bins $B_1, \dots, B_{k^*} \subset 2^{[n]}$ of minimum cardinality k^* that feasibly partition the items: for each $j = 1, \dots, k^*$,
 $\forall a \in U, \sum_{i \in B_j} a_i \leq 1$ and $\bigcup_{j \in M^*} B_j = [n]$.



Uncertainty sets to be considered

- **“Budgeted uncertainty” (Γ RBP)** Given input is $\bar{a}, \hat{a}, \Gamma$, where $\Gamma \in \mathbb{N}$,

$$U^\Gamma = \left\{ \bar{a} + \sum_{i=1}^n \delta_i \hat{a}_i e_i \mid \sum_{i=1}^n \delta_i \leq \Gamma, \delta_i \in \{0,1\} \right\}$$

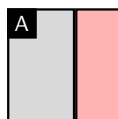
- Bertsimas & Sim (2003)

- **“Knapsack uncertainty” (Ω RBP)**. Given input is $(\bar{a}, \hat{a}, \Omega)$, where $\bar{a}, \hat{a} \in [0, 1]^n$, and $\Omega \in [0, 1]$,

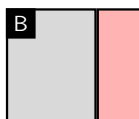
$$U^\Omega = \left\{ a \in \prod_{i \in [n]} [\bar{a}_i, \bar{a}_i + \hat{a}_i] \mid \sum_{i \in [n]} (a_i - \bar{a}_i) \leq \Omega \right\}.$$

- Bertsimas & Sim (2003)
- Gounaris, Weisemann & Floudas (2013)
- Poss (2017)

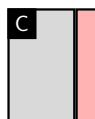
Γ -Uncertainty



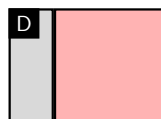
0.3 0.2



0.4 0.2

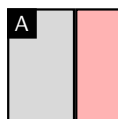


0.3 0.1

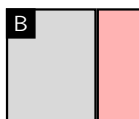


0.2 0.5

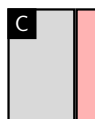
Γ -Uncertainty



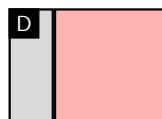
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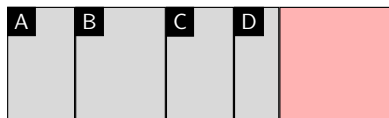
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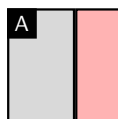
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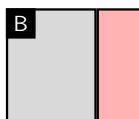
0.2 0.5

 $\Gamma = 1$, size = 1.7

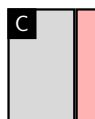
Γ -Uncertainty



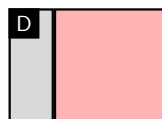
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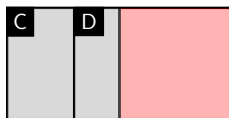
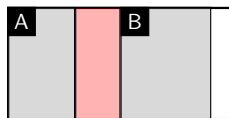
0.4 0.2



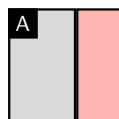
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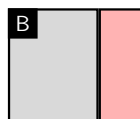
0.2 0.5



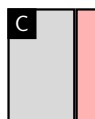
Ω -Uncertainty



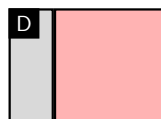
0.3 0.2



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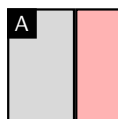


0.3 0.1

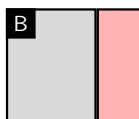


0.2 0.5

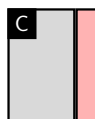
Ω -Uncertainty



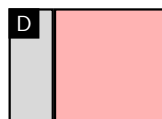
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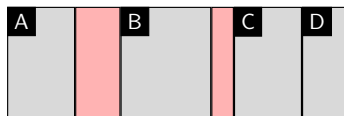
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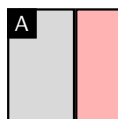
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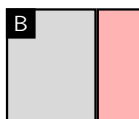
0.2 0.5

 $\Omega = 0.3$, size = 1.5

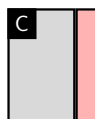
Ω -Uncertainty



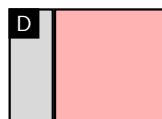
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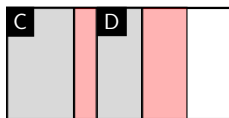
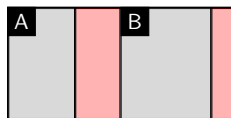
0.4 0.2



0.3 0.1



0.2 0.5



$\Omega = 0.3, 2 \text{ bins}$

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Complexity lower bound

- From standard bin packing, absolute LB $\rho \geq 3/2$ for both Γ RBP and Ω RBP.
- We now show that *unlike classical BP* Γ RBP *does not have an asymptotic FPTAS*.

Theorem

Assuming $P \neq NP$, $\rho_\infty \geq 3/2$ for Γ RBP.

Proof of lower bound $\rho_\infty \geq 3/2$

Proof.

Given a 2-partition decision problem instance consisting of rational numbers $s_1, \dots, s_N$, whose sum is 2, determine whether there is a subset whose sum is 1.

Create K Γ RBP item sets for an arbitrarily large K (so $n = NK$).

For $\delta \in (0, s_{\min})$ and each set $k = 1, \dots, K$, $\epsilon_k = \delta^k$, the nominal and deviation sizes are

$$\bar{a}_{i,k} = \epsilon_k \cdot s_i \quad \text{and} \quad \hat{a}_{i,k} = 1 - \epsilon_k.$$

Note that $\bar{a}_{i,k} > \epsilon_{k+1}$ for all $k = 1, \dots, K-1$.

Proof - cont'd

Lemma

Suppose $\Gamma = 1$. It is NP-hard to determine whether the K item sets can be packed in fewer than $3K$ bins.

Proof - cont'd

Lemma

Suppose $\Gamma = 1$. It is NP-hard to determine whether the K item sets can be packed in fewer than $3K$ bins.

Proof.

Clearly a YES-instance of partition requires $2K$ bins.

A NO-instance will require at least $3K$ bins; for any pair of items (i, k) and (i', k') of sets k and $k' > k$,

$$\bar{a}_{i,k} + \bar{a}_{i',k} + \max\{\hat{a}_{i,k} + \hat{a}_{i',k'}\} > \bar{a}_{i,k} + \hat{a}_{i',k'} > \epsilon_{k'} + (1 - \epsilon_{k'}).$$

So items of different sets k, k' cannot fit in the same bin. □



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NEXT-COVER-AND-REPACK algorithm

NEXT-COVER-AND-REPACK($\mathcal{I}$) for uncertainty set U

initialization: $k' = 0$

repeat

 Open bin $k' \leftarrow k' + 1$.

 Pack items $[n]$ according to the index order in $B_{k'}$ until
 $f_U(B_{k'}) > 1$ or all items are packed.

until *all items are packed*

Repack the last item of each bin into a new (empty) bin.

return : $B_1, B_2, \dots, B_{2k'}$

- Will show that NCR with arbitrary item ordering is not useful for approximation.
- We will consider two possible item orderings: (1) decreasing deviation-nominal size ratios and (2) decreasing deviation order.

NCR with an arbitrary order

$$n \text{ items, } \epsilon = \frac{1}{2n}$$

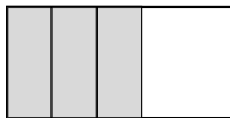
$$\Omega = 1 - \epsilon \text{ or } \Gamma = 1$$


 $2\epsilon, 0$

 $0, 1 - \epsilon$

 $2\epsilon, 0$

 $0, 1 - \epsilon$

 $2\epsilon, 0$


Optimum

NCR approximation for Ω RBP (knapsack uncertainty)

Here, we consider an ordering of the items such that

$$\frac{\hat{a}_1}{\bar{a}_1} \geq \dots \geq \frac{\hat{a}_n}{\bar{a}_n}. \quad (1)$$

Lemma

Suppose that the items are ordered according to (1). Then $k' \leq k^$.*

Theorem

If the items are ordered according to (1) then NEXT-COVER-AND-REPACK is a 2-approximation algorithm for Ω RBP.

Proof sketch.

- $G^* = \{j \in [k^*] \mid \hat{a}(B_j^*) > \Omega\}$



Proof sketch.

- $G^* = \{j \in [k^*] \mid \hat{a}(B_j^*) > \Omega\}$
- G denote the first $|G^*|$ bins opened by NCR



Proof sketch.

- $G^* = \{j \in [k^*] \mid \hat{a}(B_j^*) > \Omega\}$
- G denote the first $|G^*|$ bins opened by NCR
- $\bar{a}(B_l) > 1 - \Omega, \forall l \in G$ and $\bar{a}(B_l) \leq 1 - \Omega, \forall l \in G^*$



Proof sketch.

- $G^* = \{j \in [k^*] \mid \hat{a}(B_j^*) > \Omega\}$
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$$\Rightarrow \sum_{j \in G} \bar{a}(B_j) > \sum_{j \in G^*} \bar{a}(B_j^*)$$



Proof sketch.

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$$\Rightarrow \sum_{j \in G} \bar{a}(B_j) > \sum_{j \in G^*} \bar{a}(B_j^*)$$

$$\Rightarrow \sum_{j \in G} \hat{a}(B_j) > \sum_{j \in G^*} \hat{a}(B_j^*)$$



Proof sketch.

- $G^* = \{j \in [k^*] \mid \hat{a}(B_j^*) > \Omega\}$

- G denote the first $|G^*|$ bins opened by NCR

- $\bar{a}(B_l) > 1 - \Omega, \forall l \in G$ and $\bar{a}(B_l) \leq 1 - \Omega, \forall l \in G^*$

$$\Rightarrow \sum_{j \in G} \bar{a}(B_j) > \sum_{j \in G^*} \bar{a}(B_j^*)$$

$$\Rightarrow \sum_{j \in G} \hat{a}(B_j) > \sum_{j \in G^*} \hat{a}(B_j^*)$$

- $|[k'] \setminus G| \leq |[k^*] \setminus G^*|$



NCR approximation for Γ RBP (budgeted uncertainty)

Consider here an ordering of the items such that

$$\hat{a}_1 \geq \dots \geq \hat{a}_n. \quad (2)$$

Theorem

If items are ordered according to (2) then

NEXT-COVER-AND-REPACK is a 2Γ -approximation for Γ RBP.

Proof.

Consider $\Gamma \times \Gamma$ matrix: Letting $\mathcal{M} \subseteq \{B_1, \dots, B_{k^*}\}$ be the set of bins containing only *nondeviating or smallest deviating items* of an optimal solution $\{B_1^*, \dots, B_{k^*}^*\}$. Then,

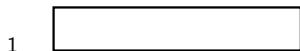
Lemma

$$|\{B_1, \dots, B_{k^*}\} \setminus \mathcal{M}| \leq (\Gamma - 1)k^*$$

Lemma

$$|\mathcal{M}| \leq k^*$$

NCR (before repacking)



⋮



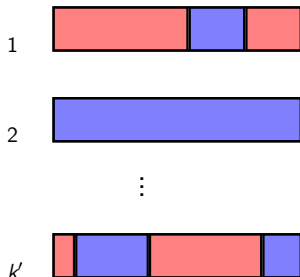
OPT



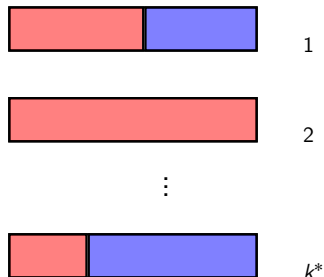
⋮



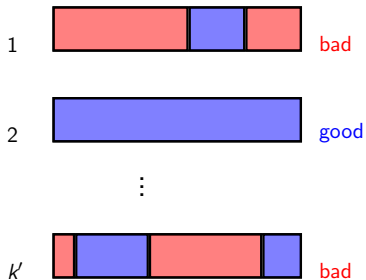
NCR (before repacking)



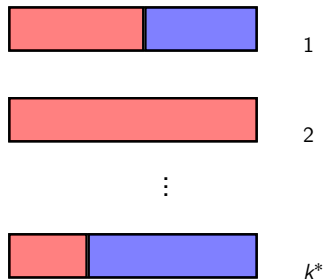
OPT



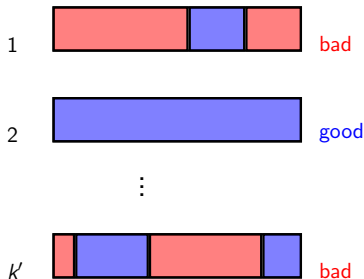
NCR (before repacking)



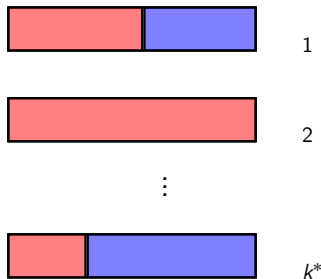
OPT



NCR (before repacking)



OPT



$$k' = \# \text{bad} + \# \text{good}$$

$$\# \text{bad} \leq (\Gamma - 1)k^*, \# \text{good} \leq k^*$$

$$k' \leq \Gamma \cdot k^* \Rightarrow \text{NCR (after repacking)} \leq 2\Gamma \cdot k^*$$

Lower bounds

Theorem

If the items are ordered according decreasing deviations order (2), then the approximation ratio of NEXT-COVER-AND-REPACK for Γ RBP is at least $2/3\Gamma$.

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Proof.

$$\begin{array}{cccccc}
 (\epsilon, 1/\Gamma - \delta_1) & (0, 1/\Gamma - \delta_1) & \dots & \dots & (0, 1/\Gamma - \delta_1) \\
 \vdots & \vdots & \ddots & \vdots & \\
 \vdots & \vdots & & \ddots & \\
 (\epsilon, 1/\Gamma - \delta_\Gamma) & (0, 1/\Gamma - \delta_\Gamma) & \dots & \dots & (0, 1/\Gamma - \delta_\Gamma)
 \end{array}$$

where $\epsilon \leq 1/\Gamma$ and $\delta_1 \leq \dots \leq \delta_\Gamma < \epsilon/\Gamma$.



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$\vdots$	$\vdots$	$\ddots$	$\vdots$	
$\vdots$	$\vdots$		$\ddots$	
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$(\epsilon, 1/\Gamma - \delta_1)$	$(0, 1/\Gamma - \delta_1)$	...	...	$(0, 1/\Gamma - \delta_1)$
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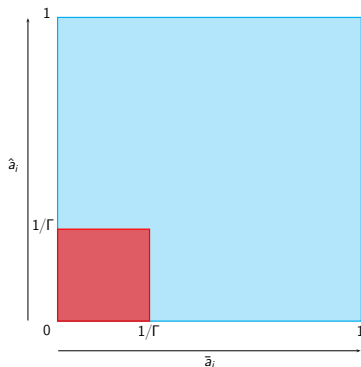
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A different construction can be used to show similar result
NEXT-COVER-AND-REPACK with the decreasing ratio order (1).

- 1 Introduction to bin packing and robust optimization
- 2 Problem definition and uncertainty sets U^{Ω} and U^{Γ}
- 3 Complexity
- 4 Next-Cover-and-Repack approximations for U^{Ω} and U^{Γ}
- 5 Constant factor approximation for U^{Γ}**
- 6 Conclusion

4.5-Approximation Algorithm for Γ -RBP

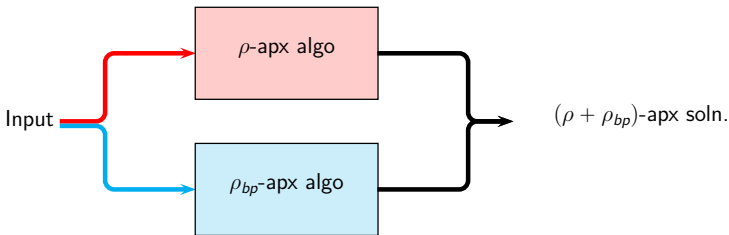


- $\Gamma/2$ small items can be packed in a bin.
- Γ RBP for large items is classical BP.

Budgeted uncertainty - constant factor approximation

Lemma

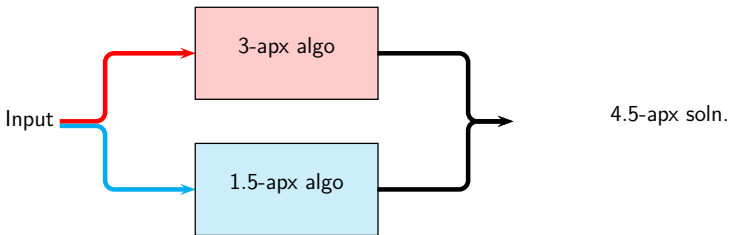
Any polynomial ρ -approximation for Γ RBP with small values (where for all i , $\bar{a}_i, \hat{a}_i \leq 1/\Gamma$) implies a polynomial $(\rho + \rho_{bp})$ -approximation for Γ RBP, where ρ_{bp} is the best known ratio of a polynomial time approximation for classical bin-packing.



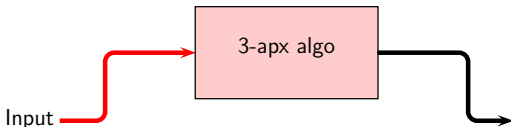
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A key observation in packing small items



Observation

Given an instance $\mathcal{I}$ to the Γ RBP with small values ($\bar{a}_i, \hat{a}_i \leq 1/\Gamma$ for all i), any subset $X \subseteq [n]$ can be packed in $\lceil \frac{|X|}{\Gamma/2} \rceil$ bins.

A key auxiliary subproblem for our algorithm:

Robust Bin Packing with Trash (Γ RBP-T)



Input: $\mathcal{I}' = (\mathcal{I}, k, q, t)$, where $\mathcal{I}$ is an instance of Γ RBP with small values, for $q, t \in [n]$, $k \in \mathbb{N}$.

Output: A feasible solution s is a partition of $\{q, \dots, n\}$ into $k + 2$ sets, $(L, B_1, \dots, B_k, T)$, satisfying

- $\tilde{f}(B_j) \leq 1$ for $j \in [k]$
- $|T| \leq t$
- $q \in B_1$

Minimize: $\bar{a}(L)$

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- $q \in B_1$

Minimize: $\bar{a}(L)$

Can show that if k^* is optimal for Γ RBP then there exists a q for which $(\mathcal{I}, k^*, q, (\Gamma - 1)k^* - q + 1)$ has $\bar{a}(L) = 0$.

3-Approximation Algorithm for Γ -RBP with small items

Γ RBP  $\bar{a}(L) = 0$ for $(\mathcal{I}, k, q, t)$ - “yes instance”

iff items $q, \dots, n$ are packed* in $\leq k$ bins with $\leq t$ items in 

Γ RBP

$k^* = OPT$

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$(\mathcal{I}, k^*, q, (\Gamma - 1)k^* - q + 1)$ is an **almost feasible** yes instance

Get a solution with $3k^*$ bins

$\Gamma k^* = t + q - 1 + k^*$ small items can be packed

in $\lceil \frac{\Gamma k^*}{\Gamma/2} \rceil = 2k^*$ bins

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DP

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trash



not packed



repacking

in $\lceil \frac{\Gamma k^*}{\Gamma/2} \rceil = 2k^*$ bins

The DP for computing an optimal almost-feasible solution for Γ RBP-T

- The DP is defined over states $(k', q', t') \in [k] \times [n] \times ((\Gamma - 1)k] \cup \{0\})$.
- The DP determines:
 - The range of items packed, the first of which q is a largest deviation item in the first bin packed.
 - The items to be trashed (largest nominal size).
 - The leftover items whose nominal size is minimized as the objective.

DP(k', q, t)

```

1  $s \leftarrow (\emptyset, \emptyset, \emptyset)$  // where  $c((\emptyset, \emptyset, \emptyset)) = \infty$ 
2 if  $k' = 1$  or  $q = n$  then
3    $B_1 \leftarrow \{q\}$ 
4    $X' \leftarrow \{q+1, \dots, n\}$ 
5    $T \leftarrow \{\min(t, n-q) \text{ largest nominal value items in } X'\}$ 
6   Pack  $X' \setminus T$  in  $B_1$  until  $\tilde{f}(B_1) > 1$  or  $X' \setminus T = \emptyset$ 
7   Pack the remaining items in  $L$ 
8   return:  $s = (L, B_1, T)$ 
9 for  $g = (q', t') \in \{q+1, \dots, n\} \times [\min(t, q' - q)]_0$  do
10    $B_1 \leftarrow \{q\}$ 
11    $X' \leftarrow \{q+1, \dots, q' - 1\}$ 
12    $T' \leftarrow \{t' \text{ largest nominal value items in } X'\}$ 
13   Pack  $X' \setminus T'$  in  $B_1$  until  $\tilde{f}(B_1) > 1$  or  $X' \setminus T' = \emptyset$ 
14   Pack the remaining items of  $X' \setminus T'$  in  $L$ 
15    $\tilde{s} = (\tilde{L}, \tilde{B}_1, \dots, \tilde{B}_{k'-1}, \tilde{T}) \leftarrow \text{DP}(k' - 1, q', t - t')$ 
16   Unpack  $\tilde{L}$  into  $B_1$  until  $\tilde{f}(B_1) > 1$  or all items of  $\tilde{L}$  are unpacked.
17   Unpack any remaining items of  $\tilde{L}$  into  $L$ .
18    $s_g \leftarrow (L, B_1, \tilde{B}_1, \dots, \tilde{B}_{k'-1}, \tilde{T} \cup T')$ 
19   if  $c(s_g) < c(s)$  then  $s \leftarrow s_g$ 
20 return:  $s$ 

```

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```

$O(n^3)$

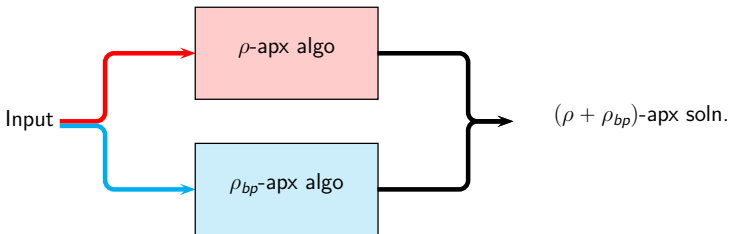
Lemma

There exists a 3-approximation for Γ RBP with small values running in $O(n^7 \log n)$.

Combining with $\frac{3}{2}$ -approximation for standard bin packing for large items:

Theorem

There exists a 4.5-approximation for Γ RBP running in $O(n^7 \log n)$.



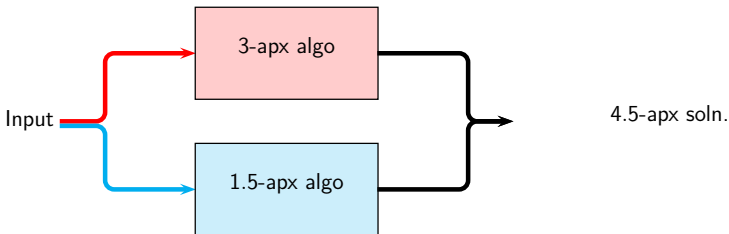
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Alternatively, an asymptotic $(4 + \epsilon)$ -approximation is also possible if “large items” are packed using Karmakar and Karp’s AFPTAS.

Conclusions

- No asymptotic PTAS for Γ RBP (in fact no asymptotic approximation less than $3/2$)
- 2-factor approximation for Ω RBP.
- $\min\{2\Gamma, 4.5\}$ -factor approximation for Γ RBP.

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- Multitype setting - constraints on assignment of items to bins

Thank you!



Marin Bougeret, Gyorgy Dosa, G. and Michael Poss.
Constant-factor approximation for robust bin-packing with budget uncertainty, revised & resubmitted, 2021.



Aniket Basu Roy, Marin Bougeret, G. and Michael Poss.
Approximating the robust bin-packing with budget uncertainty,
WADS 2019.

Questions?

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